MAN00035C

Quantitative Analysis in Business Studies

2026 – Week 1

UNIVERSITY of York
School for Business and Society

What is Quantitative Analysis?

- This course is all about data and its interpretation.
- Numbers, categories, it’s all data.
- We will look at how to **summarise** it, **describe** it, **present** it, and **analyse** it.

As we progress…

- We will look at judging whether there are connections between things.

For example:

- That taller people tend to be heavier might seem obvious, but is there a link between whether people eat breakfast and their height?

Teaching Team

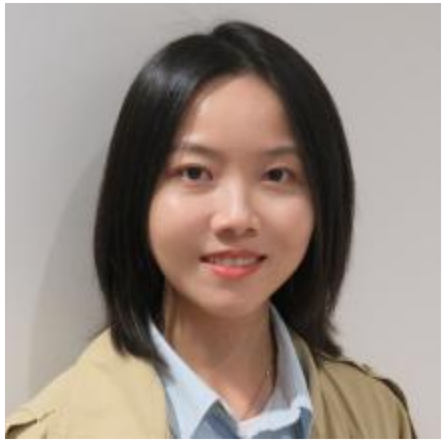
Alexandra Dias (Weeks 1-4 & 10-11)

Yuning Li (Weeks 5-9)

Seminar/Practicals Team

Kiet Duong (Weeks 2-11)

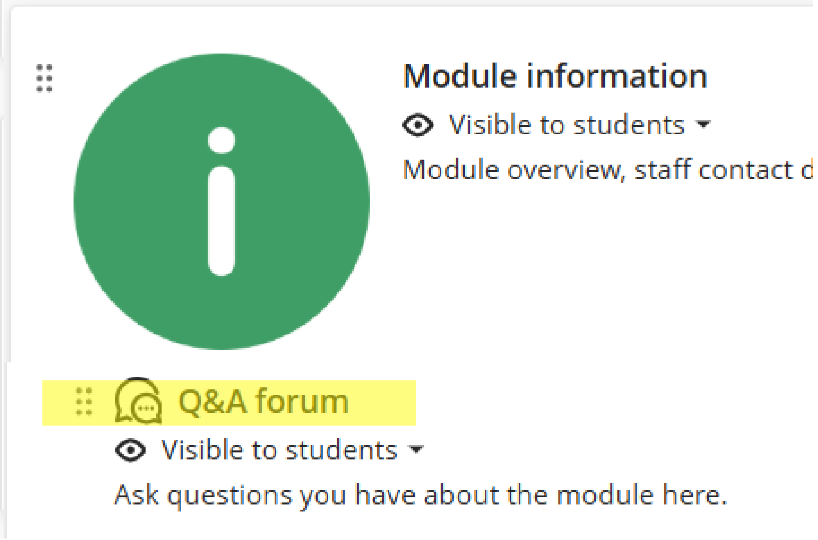
Silvia Pazzi (Weeks 2-11)



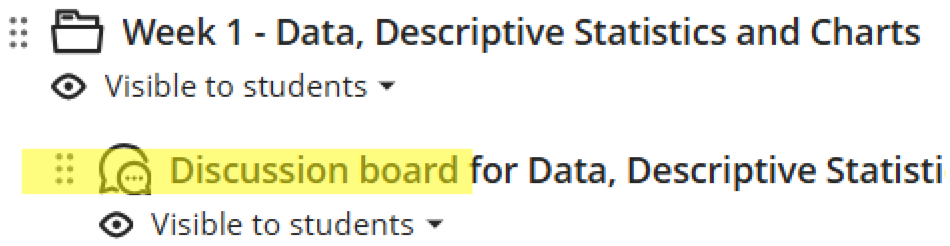
Ruoran Zhao (Weeks 2-11)

How to ask questions?

- General questions about the module:
  - Q&A Forum



- Questions on the module content:
  - Weekly Discussion Boards

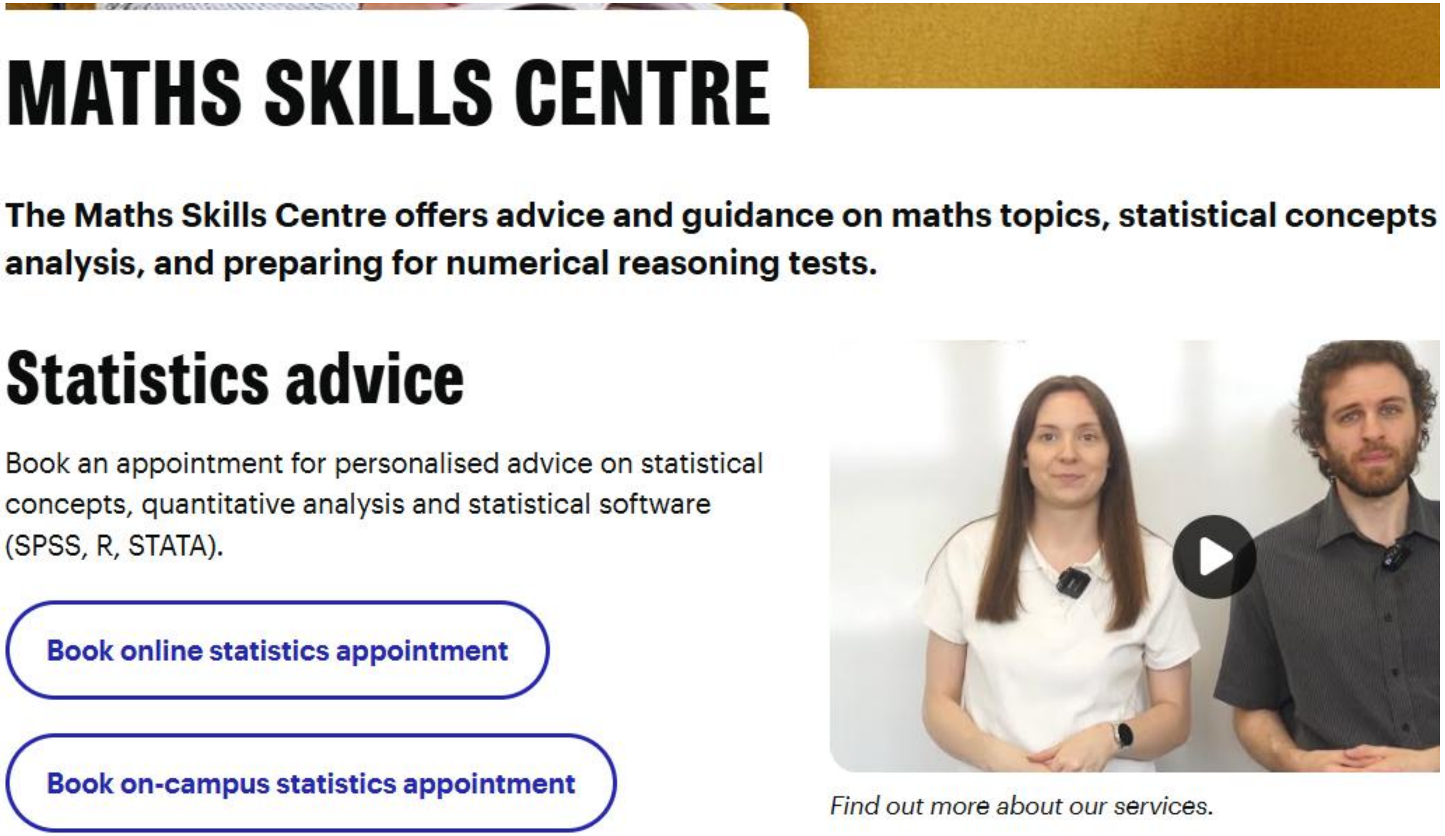


  - Office hours (see VLE module page for time and office location)
  - Email the Lecture or Seminar lecturer (email addresses on the VLE page)

Ask for help

You can get more help from The Maths Skills Centre, which offers advice and guidance on maths topics, statistical concepts and analysis, and preparing for numerical reasoning tests. For more information visit:

- https://www.york.ac.uk/students/studying/skills/maths-skills-centre



Module structure

- Each week of Semester 2:

  – **Lectures**: Always check your timetable!

    – F/100 Central Hall, Campus West - Central Hall.

  - **Seminars**: Please check your individual timetables.

- Plus, **3 weeks of workshops** in Weeks 9, 10, 11 (Please check your individual timetables)

Use the lecture slides

- The Lecture slides are available on the VLE page of the module.

- You can print them off with multiple slides on each page and space for notes. You could then use these to add notes during the lectures.

- For each lecture there is a list of readings. Please read the main text before each lecture.

Textbook(s)

- **Main text – Oakshott, L. (2021). Essential Quantitative Methods for Business, Management & Finance. 7th edition, Basingstoke: Palgrave Macmillan.**

- Other – Taylor, S. (2007). Business Statistics for Non-Mathematicians, 2nd edition. Basingstoke: Palgrave Macmillan.

- Old editions and other similar textbooks are also useful – there are many available in the library, it's worth having a look.

* Each week's reading is available on VLE

Textbook(s)

- Where to find them?
  - VLE module page under Reading Lists:

Reading-Lists
Visible to students
Leganto Reading Lists

Course reference (6)

Essential quantitative methods for business, management, and finance
Book | Oakshott, Les, Seventh edition., London :, Red Globe Press, 2020., Total pages xxvii, 427 pages :
Note: You will have to register with Kortext the first time you use one of their ebooks. Institution = University of York, and use your York logins.
View online · Available at University Library Morrell - Ordinary: G 5.8 OAK

Business statistics for non-mathematicians
Book | Taylor, Sonia., 2nd ed., Basingstoke :, Palgrave Macmillan, 2007., Total pages xvi, 368 p. :
Available at University Library Morrell - Ordinary: SF 1.G TAY

Additional reading

- Most lectures have other associated pieces of reading and miscellanea, these are all linked from the VLE.

- These are not compulsory, but all are useful to develop your intuition and thinking around the course material.

Every week: Formative Quiz

- For each of the topics, there is a formative quiz available on VLE.
- Please try to solve it before each seminar.

Week 1 - Data, Descriptive Statistics and Charts

Data, Descriptive Statistics and Charts - Formative Quiz

No due date

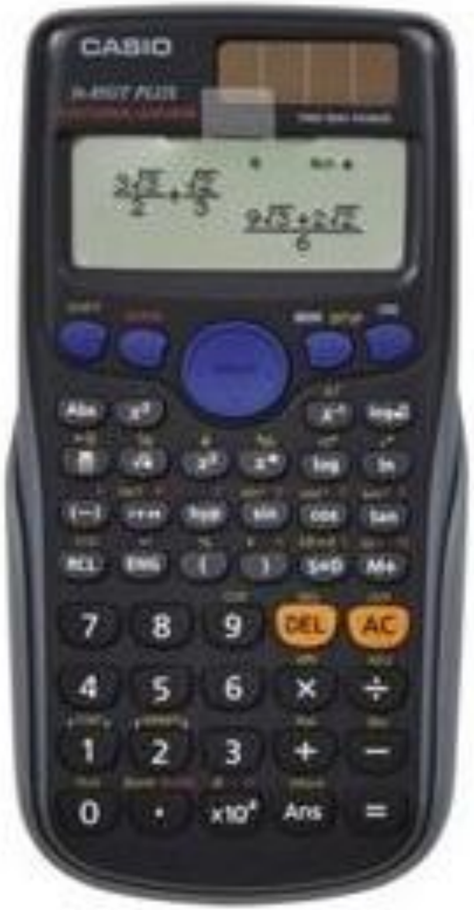
A selection of questions about the topic Measures and Charts for you to practise on. Unless otherwise instructed, and where applicable, please provide answers to four (4) decimal places. If asked, please provide answers for Coefficient of Variation as a proportion and not a percentage. Formative tests do not count towards your final course grade. You can do it many times, and generally will get different questions each time.)

Discussion board for Data, Descriptive Statistics and Charts

Calculators – you will need one

- **‘Where they are permitted in exams, calculators will now be provided by the Exams Office’** [University calculator policy for exams](https://tinyurl.com/2ubh56bz)*.

- It is worthwhile getting one of these to familiarize yourself with its functionality:
  - Casio fx-85GT Plus
  - Casio fx-85GTX Classwiz
- Try it out at the Library



* https://tinyurl.com/2ubh56bz

Assessment

This module will be assessed with two components:

1. 1200 word assignment (report/course work), due in **Week 11** (date to be announced soon on the VLE site) counting for **50%** of the module mark. The assignment will be released to you at **the end of Week 4**. The assignment is to be your own work (do not approach the teaching staff for answers).

2. A closed exam due in the Semester 2 **Common Assessment Period** (date to be announced later on the VLE site) counting for **50%** of the module mark.

Honour Code

This course requires you to prepare and submit work that is not directly supervised. You are on your honour not to act in any way that would give you an unfair advantage. That includes using other peoples work as your own and/or providing other people with answers to submittal work. You are also on your honour to report any student who so acts in a dishonourable manner.

You need to

- Attend lectures and seminars (attendance is recorded)
- Read the lecture slides in advance
- Do the formative quizzes
- Try to work out things you don’t understand
- Ask for help when you need it
- Stay up to date - Do not leave things for too late
- Ask for help when you need it
- Ask for help when you need it, and
- Ask for help when you need it ☺

Now we can start

Data, Descriptive Statistics and Charts

1. Describing Data

We might want to know:

- What **types** of data are these?
- Where do the data observations **concentrate**?
- How **spread** out are they? Measures of **dispersion** or **spread**
- How do they **compare** to other data?
- How can we **picture** it?

- **(!) The answer to the first question affects how we answer the rest.**

2. Inference from Data

Then, we might want to know:

- Is this what we expect to see?
- How are the variables **associated** with each other?
- What effect does changing one variable have on another?

Example dataset:

All 159 students enrolled in a given module

- To provide greater consistency throughout the course, we will focus on the same dataset most of the time, and examples will often be drawn from it.

- This data set is available on the VLE module page, in Excel and SPSS formats.

1	Gender	Height (m)	Weight (kg)	Height Guess	Weight Guess	Breakfast?	College	Confident With Numbers
2	Female	1.35	23	1.74	59	Yes	Goodricke	Agree
3	Female	1.45	34	1.83	80	Yes	Constantine	Disagree
4	Female	1.47	36	1.7	56	Yes	Constantine	Disagree
5	Female	1.47	25	1.58	72	Yes	Constantine	Disagree
6	Female	1.48	37	1.8	73	No	Goodricke	Disagree
7	Female	1.5	29	1.72	67	Yes	Constantine	Agree
8	Female	1.52	37	1.81	80	Yes	Goodricke	Neither Agree nor Disagree
9	Female	1.52	40	1.53	59	Yes	Goodricke	Agree
10	Female	1.52	46	1.52	77	Yes	Goodricke	Agree
11	Female	1.52	35	1.8	73	Yes	Vanburgh	Agree
12	Female	1.53	39	1.71	56	Yes	Langwith	Neither Agree nor Disagree

Dataset: 8 variables with 159 observations

	A	B	C	D	E	F	G	H
1	Gender	Height (m)	Weight (kg)	Height Guess	Weight Guess	Breakfast?	College	Confident With Numbers
2	Female	1.35	23	1.74	59	Yes	Goodricke	Agree
3	Female	1.45	34	1.83	80	Yes	Constantine	Disagree
4	Female	1.47	36	1.7	56	Yes	Constantine	Disagree
5	Female	1.47	25	1.58	72	Yes	Constantine	Disagree
6	Female	1.48	37	1.8	73	No	Goodricke	Disagree
7	Female	1.5	29	1.72	67	Yes	Constantine	Agree
8	Female	1.52	37	1.81	80	Yes	Goodricke	Neither Agree nor Disagree
9	Female	1.52	40	1.53	59	Yes	Goodricke	Agree
10	Female	1.52	46	1.52	77	Yes	Goodricke	Agree
11	Female	1.52	35	1.8	73	Yes	Vanburgh	Agree
154	Male	1.93	96	1.72	69	No	Constantine	Neither Agree nor Disagree
155	Male	1.93	97	1.86	59	Yes	Goodricke	Agree
156	Male	1.93	93	1.83	66	Yes	Langwith	Agree
157	Male	1.95	92	1.76	71	Yes	James	Neither Agree nor Disagree
158	Male	1.95	92	1.75	48	No	Halifax	Agree
159	Male	2.02	113	1.72	77	Yes	Goodricke	Strongly Agree
160	Male	2.06	107	1.69	67	Yes	Langwith	Agree
161								

Data Types

1. **Categorical**
   – **Nominal** – data that usually does not have a numerical value and can only be placed in a suitable category. Categories have **no order**.
   – **Ordinal** – data that usually does not have a numerical value but categories can be arranged in some **meaningful order**
   – **Dichotomous** - categorical variables (nominal or ordinal) with two categories
2. **Scalar**
   – **Continuous** – measured on a scale, e.g. weight or length
   – **Discrete** – data that takes on whole values, usually obtained by counting - e.g. number of defective items.

Classify each variable in these data:

Categorical:

- nominal
- ordinal
- dichotomous

Scalar:

- continuous
- Discrete

Gender	Height (m)	Weight (kg)	Height Guess	Weight Guess	Breakfast?	College	Confident With Numbers
Female	1.52	46	1.52	77	Yes	Goodricke	Agree
Male	1.72	76	1.52	66	Yes	Derwent	Agree
Female	1.52	40	1.53	59	Yes	Goodricke	Agree
Male	1.65	61	1.54	57	No	Langwith	Disagree
Male	1.8	72	1.54	77	Yes	Langwith	Agree
Female	1.54	45	1.57	61	Yes	Derwent	Disagree
Male	1.69	69	1.57	70	Yes	Derwent	Agree
Female	1.47	25	1.58	72	Yes	Constantine	Disagree
Female	1.83	69	1.58	61	Yes	James	Strongly Agree
Female	1.81	61	1.6	52	Yes	Constantine	Strongly Agree
Female	1.7	68	1.61	74	Yes	Derwent	Agree
Male	1.9	84	1.61	69	Yes	James	Agree
Female	1.57	49	1.62	64	Yes	Constantine	Agree
Female	1.65	44	1.62	70	Yes	Constantine	Strongly Agree
Female	1.71	56	1.62	74	Yes	Langwith	Neither Agree nor Disagree
Female	1.72	56	1.62	77	Yes	Constantine	Neither Agree nor Disagree
Female	1.75	71	1.62	88	Yes	Langwith	Neither Agree nor Disagree
Male	1.92	83	1.62	87	No	James	Disagree
Female	1.65	48	1.63	85	Yes	Langwith	Neither Agree nor Disagree
Male	1.71	75	1.63	70	Yes	Langwith	Neither Agree nor Disagree

Descriptive statistics

1. **Measures of tendency or centrality**
   - **Mode** – the category or value most frequently observed in the data
   - **Median** – the category or value in the middle observation after ordering the data (not suitable for nominal data)
   - **Mean or average** – sum of all the observations divided by the number of observations (only suitable for scalar data)
2. **Measures of dispersion or spread**
   - **Number of categories** – for nominal or ordinal data
   - **Range, variance, standard deviation, interquartile range** – all these only for continuous or discrete data (definitions later)

Data Types – Nominal

- **Frequency**
  - How many observations in each category
- **Percent**
  - Number of observations in each category divided by the total number of observations

Statistics

Gender

N	Valid	159
	Missing	0

Gender

		Frequency	Percent
Valid	Male	64	40.3
	Female	95	59.7
	Total	159	100.0

Data Types – Nominal

- **Measure of tendency: mode**
  - Category with the largest number of observations

- **Measure of dispersion**
  - Number of categories

Statistics

Gender

N	Valid	159
	Missing	0

Gender

		Frequency	Percent
Valid	Male	64	40.3
	Female	95	59.7
	Total	159	100.0

Data Types – Nominal

- **Measure of tendency: mode**
  - Category with the largest number of observations

- **Measure of dispersion**
  - Number of categories

Statistics

Gender

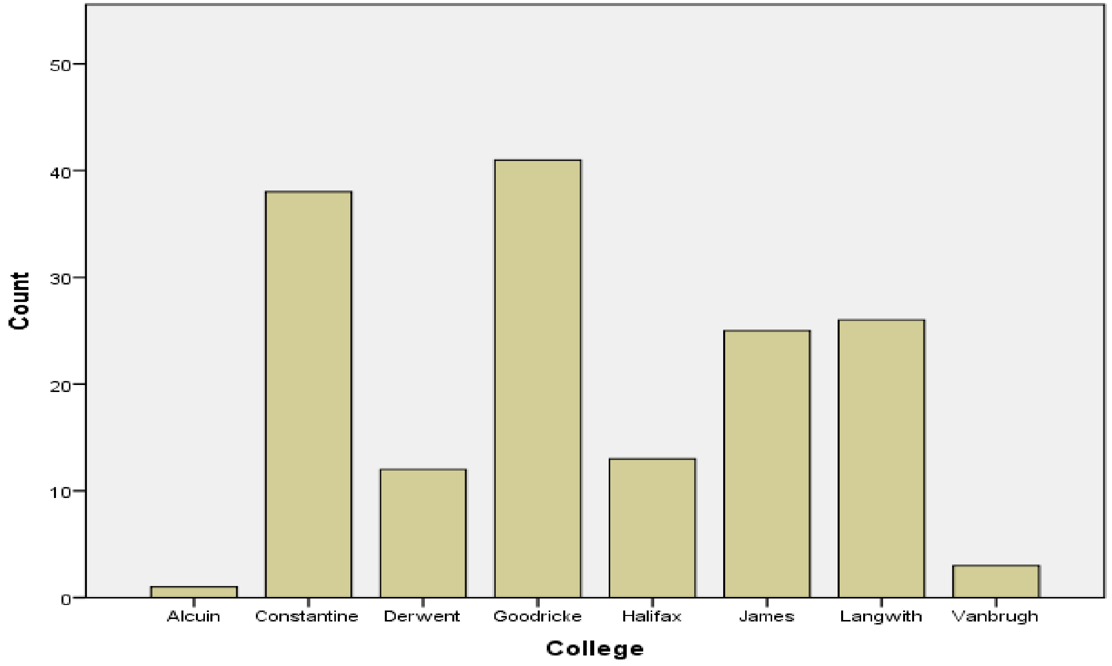
N	Valid	159
	Missing	0

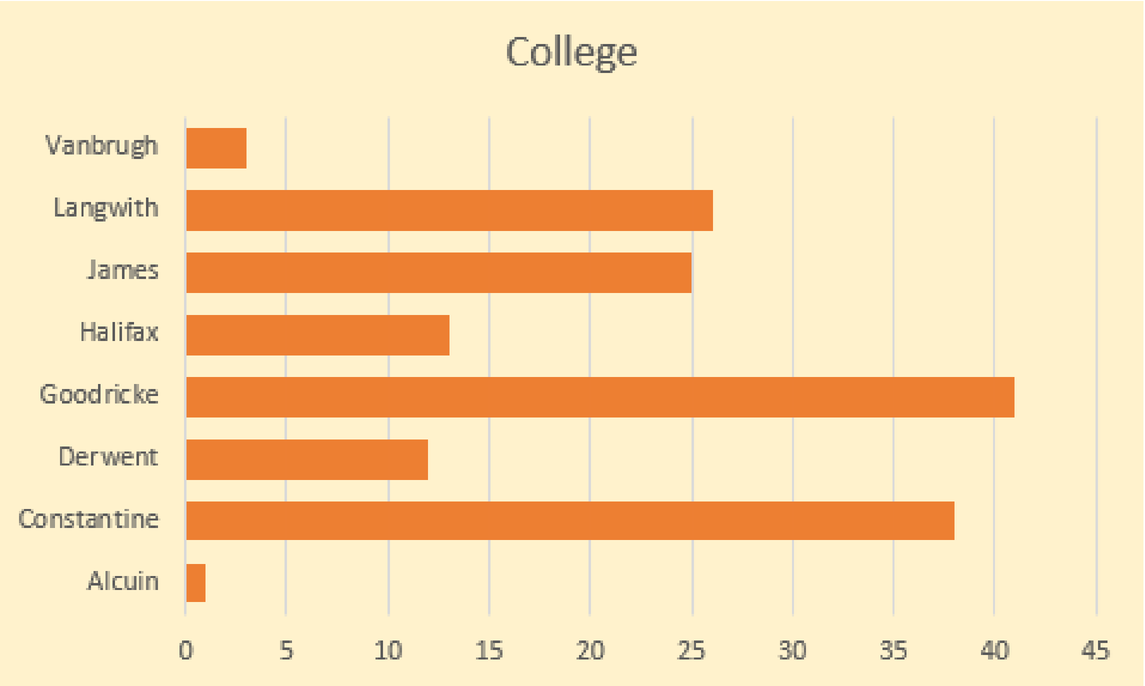
College

		Frequency	Percent
Valid	Alcuin	1	.6
	Constantine	38	23.9
	Derwent	12	7.5
	Goodricke	41	25.8
	Halifax	13	8.2
	James	25	15.7
	Langwith	26	16.4
	Vanbrugh	3	1.9
	Total	159	100.0

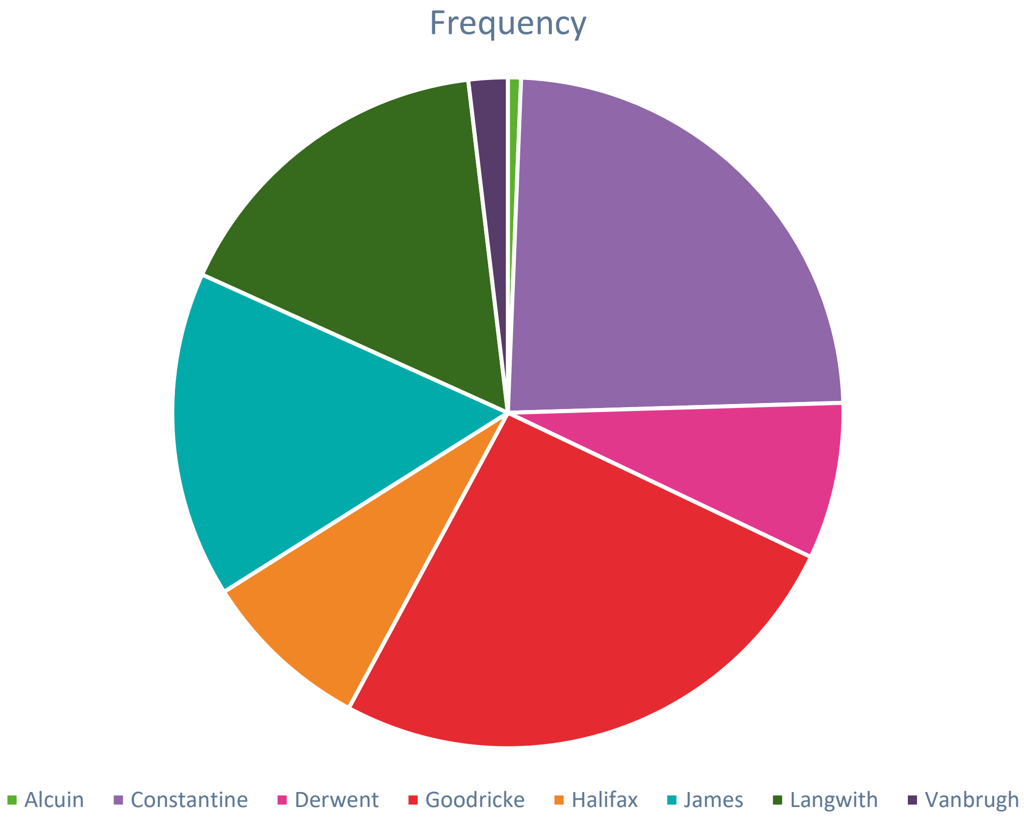
Data Types – Nominal

- **Bar Charts:**
  - show how many categories we have
  - show the mode (=> the longest one)





Data Types – Nominal



Pie Chart:

- Shows how many categories
- Helps compare proportions

Data Types – Ordinal

- In our dataset, only **Confidence With Numbers** is ordinal.

- The data includes the idea of **"order".**
  - **Strongly disagree; Disagree; Neither…; Agree; Strongly agree**

- Because it is **categorical, a bar chart is still generally the best.**

Gender	Height (m)	Weight (kg)	Height Guess	Weight Guess	Breakfast?	College	Confident With Numbers
Female	1.52	46	1.52	77	Yes	Goodricke	Agree
Male	1.72	76	1.52	66	Yes	Derwent	Agree
Female	1.52	40	1.53	59	Yes	Goodricke	Agree
Male	1.65	61	1.54	57	No	Langwith	Disagree
Male	1.8	72	1.54	77	Yes	Langwith	Agree

Gender	Height (m)	Weight (kg)	Height Guess	Weight Guess	Breakfast?	College	Confident With Numbers
Female	1.52	46	1.52	77	Yes	Goodricke	Agree
Male	1.72	76	1.52	66	Yes	Derwent	Agree
Female	1.52	40	1.53	59	Yes	Goodricke	Agree
Male	1.65	61	1.54	57	No	Langwith	Disagree
Male	1.8	72	1.54	77	Yes	Langwith	Agree
Female	1.54	45	1.57	61	Yes	Derwent	Disagree
Male	1.69	69	1.57	70	Yes	Derwent	Agree
Female	1.47	25	1.58	72	Yes	Constantine	Disagree
Female	1.83	69	1.58	61	Yes	James	Strongly Agree
Female	1.81	61	1.6	52	Yes	Constantine	Strongly Agree
Female	1.7	68	1.61	74	Yes	Derwent	Agree
Male	1.9	84	1.61	69	Yes	James	Agree
Female	1.57	49	1.62	64	Yes	Constantine	Agree
Female	1.65	44	1.62	70	Yes	Constantine	Strongly Agree
Female	1.71	56	1.62	74	Yes	Langwith	Neither Agree nor Disagree
Female	1.72	56	1.62	77	Yes	Constantine	Neither Agree nor Disagree
Female	1.75	71	1.62	88	Yes	Langwith	Neither Agree nor Disagree
Male	1.92	83	1.62	87	No	James	Disagree
Female	1.65	48	1.63	85	Yes	Langwith	Neither Agree nor Disagree
Male	1.71	75	1.63	70	Yes	Langwith	Neither Agree nor Disagree

Data Types – Ordinal

- **Measure of tendency: mode**
  - Mode: Category with the largest number of observations
  - Median: the category or value in the middle observation after ordering the data

- **Measure of dispersion**
  - Number of categories

Statistics

I am confident with numbers

N	Valid	159
	Missing	0

I am confident with numbers

		Frequency	Percent
Valid	Strongly Disagree	2	1.3
	Disagree	29	18.2
	Neither Agree nor Disagree	54	34.0
	Agree	51	32.1
	Strongly Agree	23	14.5
	Total	159	100.0

Data Types – Ordinal

- Because we have the ordering, we can pick out the value that is in the middle, **the median.**
- 159 datapoints means the middle one is the $(\frac{159+1}{2})^{\wedge}th$ one, i.e. the 80th.
- Median of the sample is **‘Neither Agree nor Disagree’**.

A	B	C	D	E	F	G	H	I
Gender	Height	Weight	Height	Weight	Breakfa	College	Confident With Numbers	Observation
Male	1.83	83	1.74	74	Yes	Goodricke	Neither Agree nor Disagree	78
Male	1.85	85	1.84	72	No	Goodricke	Neither Agree nor Disagree	79
Male	1.86	85	1.79	59	Yes	Goodricke	Neither Agree nor Disagree	80
Male	1.89	85	1.78	56	No	Constantine	Neither Agree nor Disagree	81
Male	1.9	91	1.69	72	Yes	James	Neither Agree nor Disagree	82
Female	1.91	77	1.67	67	Yes	Langwith	Neither Agree nor Disagree	83
Male	1.93	96	1.72	69	No	Constantine	Neither Agree nor Disagree	84
Male	1.95	92	1.76	71	Yes	James	Neither Agree nor Disagree	85
Female	1.35	23	1.74	59	Yes	Goodricke	Agree	86

- Note that accurate identification of the median in ordinal data is only possible if we have an odd number of data points.

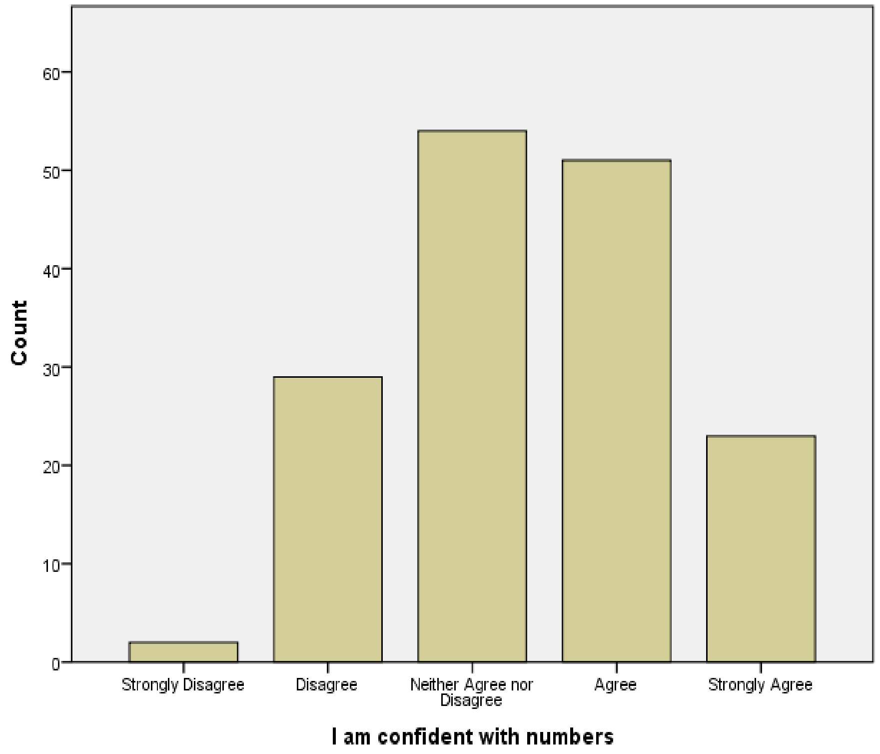
Data Types – Ordinal

- **Charts:**

- The data includes the idea of **"order".**

- Because it is categorical, a bar chart is still generally the best.

Note the order of the categories



Data Types – Scalar

- Height, Weight, and the two "guess" variables are all **Scalar** and **continuous**.
- Scalar adds the **idea of distance**, not just order.

Gender	Height (m)	Weight (kg)	Height Guess	Weight Guess	Breakfast?	College	Confident With Numbers
Female	1.52	46	1.52	77	Yes	Goodricke	Agree
Male	1.72	76	1.52	66	Yes	Derwent	Agree
Female	1.52	40	1.53	59	Yes	Goodricke	Agree
Male	1.65	61	1.54	57	No	Langwith	Disagree
Male	1.8	72	1.54	77	Yes	Langwith	Agree

Descriptive statistics for scalar data

1. **Measures of centrality**
   - **Mode** – the value most frequently observed value in discrete data
   - **Median** – the value in the middle observation after ordering the data
   - **Mean or average** – sum of all the observations divided by the number of observations

2. **Measures of dispersion or spread**
   - **Range, variance, standard deviation, interquartile range** – definitions in the following

Descriptive statistics for scalar data

Median – the value in the middle observation after ordering the data

- If the number of observations is **odd**:

  Median = (n+1)/2*th* observation

- Example with n=11 observations: Median = 6th observation = 72

Median

48 52 57 61 64 72 76 77 81 85 88

Descriptive statistics for scalar data

Median – the value in the middle observation after ordering the data

- If the number of observations is **even**:

  Median = average between n/2*th* and n/2+1*th* observations

- Example with n=10 observations:

  Median = (64+72)/2 = 68

Median

48	52	57	61	64	72	76	77	81	85

Descriptive statistics: Height data

1. **Measures of centrality**
   - **Median** – the value in the middle observation after ordering the data

A	B	C	D	E	F	G	H	I
Gender	Height	Weight	Height	Weight	Breakfa	College	Confident With Numbers	Observation
Male	1.7	68	1.71	66	No	James	Neither Agree nor Disagree	78
Male	1.7	58	1.66	77	No	James	Neither Agree nor Disagree	79
Female	1.7	68	1.61	74	Yes	Derwent	Agree	80
Female	1.7	56	1.76	67	Yes	Constantine	Agree	81
Female	1.7	57	1.73	69	Yes	Goodricke	Strongly Agree	82
Male	1.71	75	1.63	70	Yes	Langwith	Neither Agree nor Disagree	83
Female	1.71	56	1.62	74	Yes	Langwith	Neither Agree nor Disagree	84

Descriptive statistics: Height data

1. **Measures of centrality**
   - **Median** – the value in the middle observation after ordering the data
   - **Mean or average** – sum of all the observations divided by the number of observations

Height (m)	
Mean	1.705974843
Median	**1.7**

Important

- **Mean or average** – very intuitive; can be affected by very large observations
- **Median** – more robust measure of centrality than the mean

Descriptive statistics for scalar data

2. Measures of dispersion or spread

– **Range:** the highest value **(Maximum)** minus the lowest value **(Minimum)**

Minimum	1.35
Maximum	2.06
Range	.71

Data Types – Scalar

- We call the usual **average** **or** **mean** for scalar data by the Greek letter mu: **μ**

  - **μ** is the sum of all the values, divided by the total number of observations (n)

- The idea of **distance** is summarized in the measure **standard deviation,** sigma (**σ**).

  - **Standard deviation, σ** $= \sqrt{\frac{\sum_i (x_i - \mu)^2}{n}}$
  - **The standard deviation σ is the distance of data from the mean**, it does not matter if the value of the observation is above or below the mean because of the squaring. **Distance matters.**

💡 Sometimes the variance is more useful than the standard deviation.
The **variance** is the standard deviation squared, **σ^2**.

Interquartile range (IQR)

- The **IQR describes the middle 50% of values when ordered from lowest to highest**.
- To find the interquartile range (IQR), **first find the median (middle value)** of the **lower and upper half of the data**.
- These values are **1st** (or lower) **quartile (Q1)** and **3rd** (or upper) **quartile (Q3).**
- The **IQR is the difference between Q3 and Q1.**
- Note: the median is the second quartile, Q2

Interquartile range (IQR)

- **IQR:** To calculate the IQR, we set the $\frac{n+1}{4}$ th data value to be **the lower quartile (LQ)**, and the $3\left(\frac{n+1}{4}\right)$ th one as the **upper quartile (UQ)**. Then, IQR = UQ – LQ.

- **Example of IQR:** for the 159 datapoints, we'd be using the $\frac{159+1}{4}^{th} = 40^{th}$ one and the $3\left(\frac{159+1}{4}\right)^{th} = 120^{th}$ one. This gives 1.80m – 1.61m = 0.19m as the IQR.

- This rule works well when **(n+1)/4 is integer or whole number**

Interquartile range: when n is even

- Example when n = 12

  Q1 = (61+64)/2 = 62.5, Q3 = (85+89)/2 = 87 and IQR = 87-62.5 = 24.5

Q1 Q3

48 52 61 64 72 76 | 77 81 85 89 90 93

- Example n = 14

  Q1 = 64, Q3 = 90 and IQR = 90-64 = 26

Q1 Q3

45 52 61 64 72 76 77 | 81 85 89 90 93 96 98

Interquartile range: when n is odd

- Example n = 13
  Q1 = (64+72)/2=68, Q3 = (90+93)/2=91.5 and IQR = 91.5-68 = 23.5

Q1 Median Q3

52 61 64 72 76 77 81 85 89 90 93 96 98

- Example when n = 11 [here (n+1)/4 = 3 is integer!]
  Q1 = 57, Q3 = 81 and IQR = 81-57 = 24

Q1 Median Q3

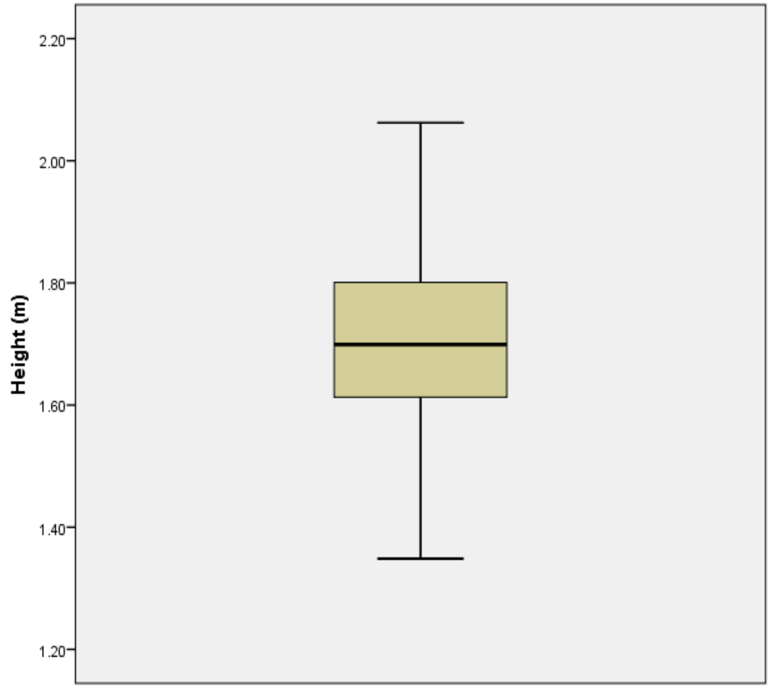
48 52 57 61 64 72 76 77 81 85 88

Data Types – Scalar

Boxplot: a chart displaying five descriptive statistics of a dataset variable (e.g. Height)

Are you able to identify in the boxplot?

1. the **minimum** (smallest observation)
2. the **lower quartile** or **first quartile**
3. the **median** (the middle value)
4. the **upper quartile** or **third quartile**
5. the **maximum** (largest observation)



The median and associated other points (the maximum, minimum, lower quartile and upper quartile) are displayed on a **boxplot**. (a box and whisker plot)

Data Types – Scalar

- SPSS can generate a table showing a range of **summary statistics** for a single scalar variable.
- We will see how to work with it at the practicals.
- **Use your calculator**; it will save you time and increase your accuracy! (you must know the formulas, though).
- Or use **Excel**.

Descriptives

			Statistic
Height (m)	Mean		1.7061
	95% Confidence Interval for Mean	Lower Bound	1.6863
		Upper Bound	1.7260
	5% Trimmed Mean		1.7051
	Median		1.6991
	Variance		.016
	Std. Deviation		.12689
	Minimum		1.35
	Maximum		2.06
	Range		.71
	Interquartile Range		.19
	Skewness		.161
	Kurtosis		-.248

When context matters

- All descriptive statistics until now are **absolute measures**
- Not always appropriate

Example: What is more severe?

- A £1,000,000 loss for a large oil company, OR
- A £100,000 loss for a small family business ?

Now, let's introduce two useful **relative measures**:

- **coefficient of variation**
- the idea of **standardising data**

Coefficient of Variation

The **coefficient of variation**: **CV** $= \frac{\sigma}{\mu}$

- is the **standard deviation** divided by the **mean**
- is higher the greater the level of **dispersion around the mean** (lower accuracy)
- can be **positive** or **negative**.

Standardising

- **Which is biggest: 10 meters, 10 kilograms, or 10 hours?**

- How to **compare** things **with different units**? We want to be able to compare like with like.

➢ **Standardisation**:

For a data point x in a set of data with a mean μ and std. dev. σ, we can standardise x to z by:

$$z = \frac{x - \mu}{\sigma}$$

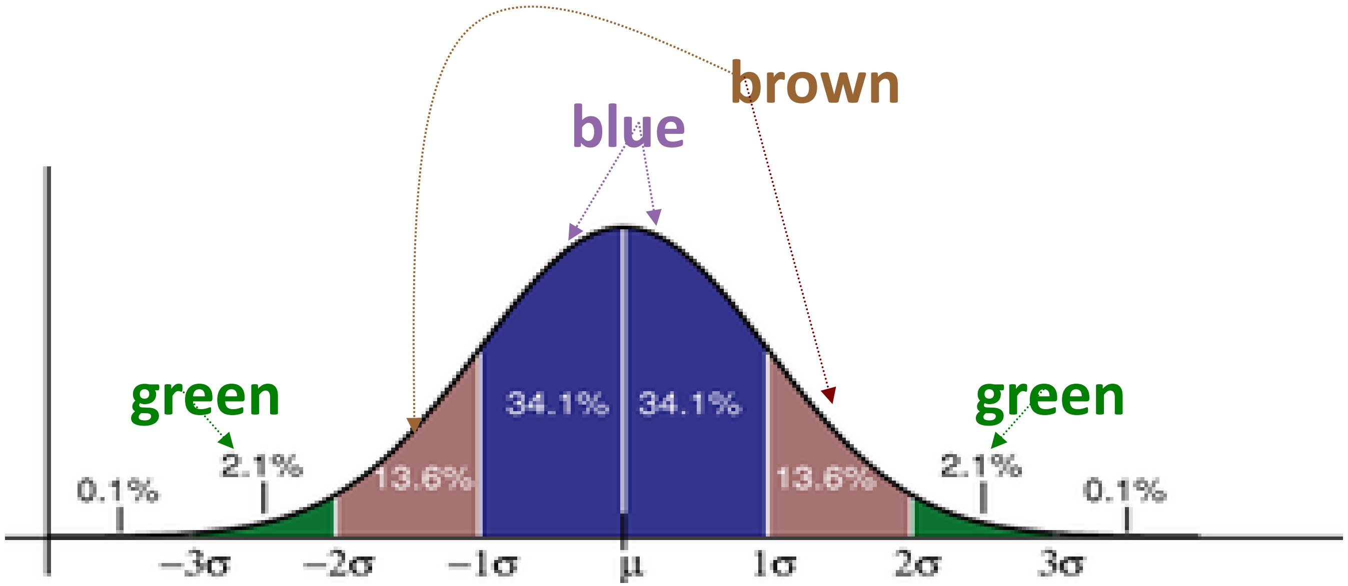
- **z** measures (or **z-score**): how many standard deviations is a data point x from the mean.

Example

STANDARDIZE Function

Name	Points	Z-Score
Edward	90	=STANDARDIZE(C5,C15,C16)
Nelissa	95	0.7438
William	92	0.24793
Tina	93	0.41322
Smith	98	1.23967
Arik	95	0.7438
Ryan	98	1.23967
Ronald	90	-0.0826

Average	90.5
STDEV.P	6.05



- Source: https://corporatefinanceinstitute.com/resources/excel/z-score-standardize-function/

Lastly....

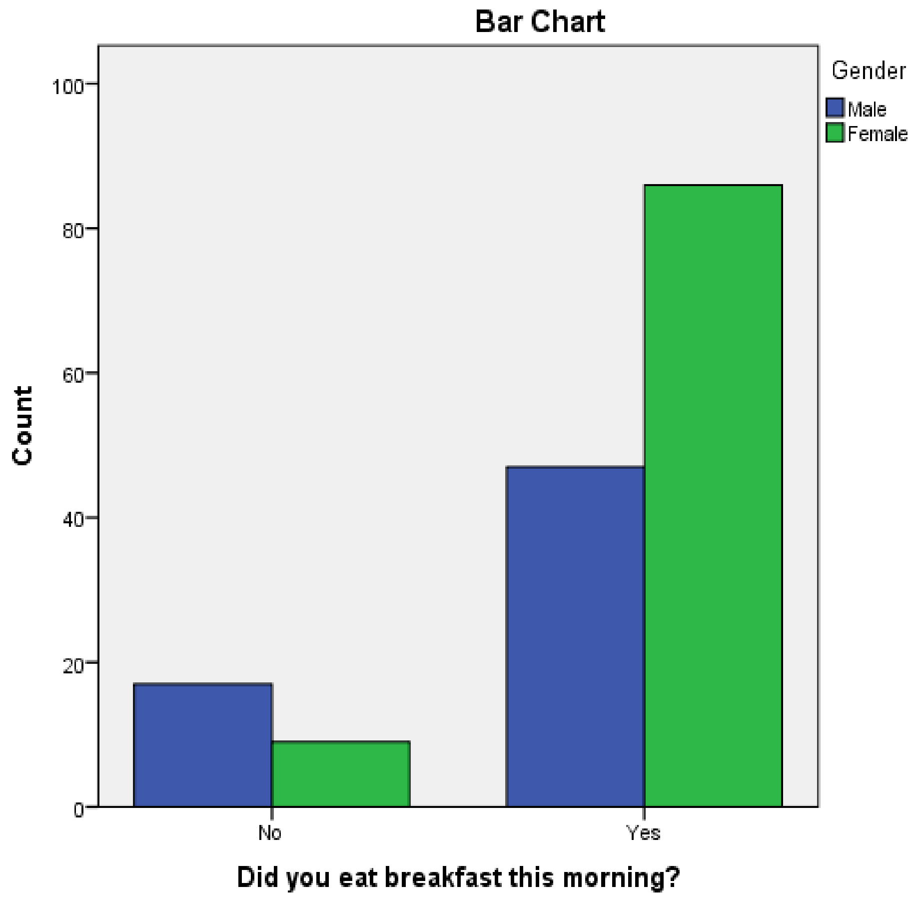
- We can combine variables to extract specific conclusions…

- i.e., **categorical** related to another **categorical** variable

Gender * Did you eat breakfast this morning? Crosstabulation

Count

		Did you eat breakfast this morning? No	Did you eat breakfast this morning? Yes	Total
Gender	Male	17	47	64
	Female	9	86	95
Total		26	133	159

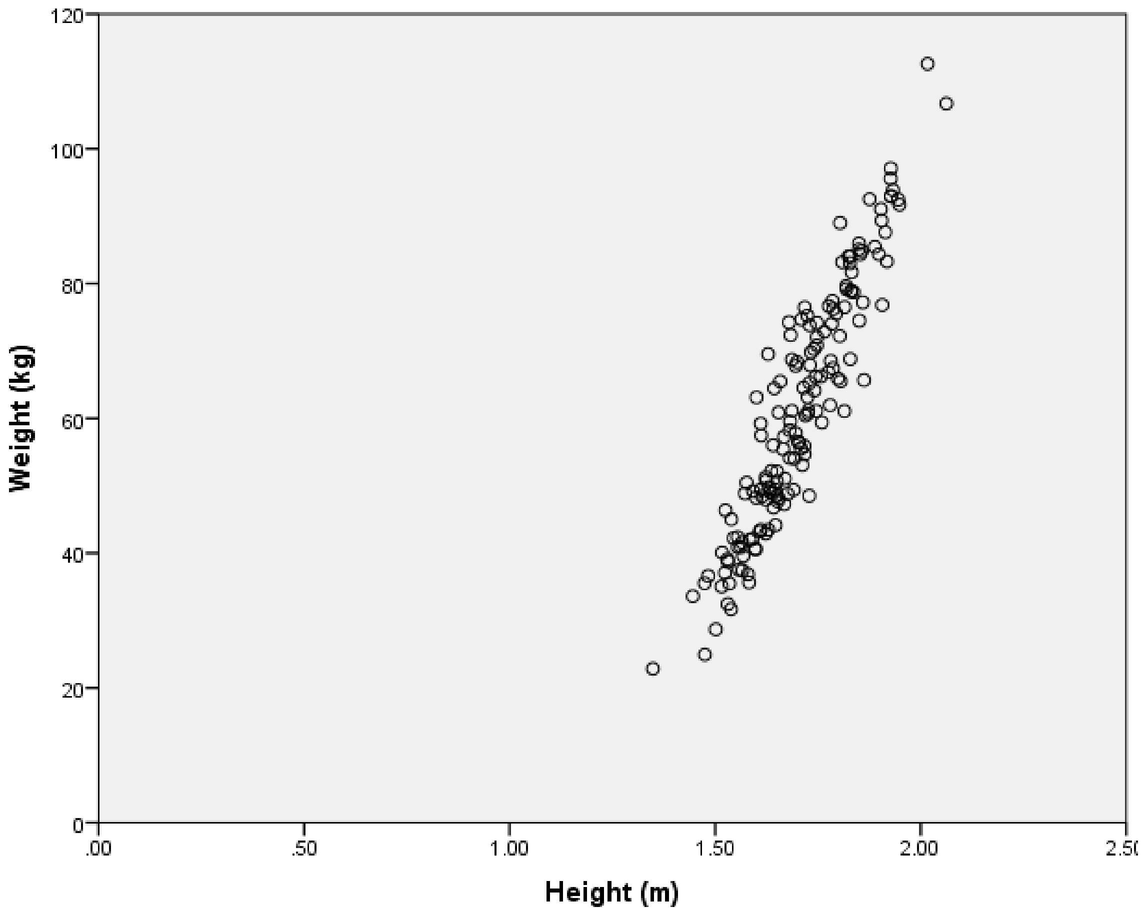


Lastly….

- We can combine variables to extract specific conclusions…

- i.e., **Scalar** related to a **scalar**

- We will return to these graphs in the practicals.



Summary:

- Data types

- Descriptive statistics
  - Measures of tendency or centrality
  - Measures of dispersion or spread:

- Absolute measures
- Relative measures

Key points

- The **mean, median** and **mode** are **measures of centrality**

  The median is less affected by unusual large observations than the mean

- To describe a set of data fully, a **measure of spread** is required. The **range** is the simplest measure, but a better measure is the **interquartile range (IQR).**
- **IQR** uses the middle 50 percent of the data and is therefore **less influenced by extreme values.**

- Another **measure of spread** is **variance**. The square root of variance is the **standard deviation**, which is in the <u>same units as the data</u>.

- When comparing the spread of two or more variables, it is useful to compare the **coefficients of variation** for each as these take into account differences in the mean.
- **Standardization** allows to compare variables with different units.

Next topic

We will consider:

2. Probability

- Basic definition
- Conditional probability and independence
- Bayes’ Law

Recommended reading:

- This week’s lectures: Oakshott Ch5 and Taylor Ch11.
- Next week’s lectures: Oakshott Ch6